

General Industries Hub

Enterprise Inventory Management

Product Manual & User Catalogue

Version 2.0 - Generated 2026-06-12

Confidential - for authorized personnel only

GI Hub - Streamlit + SQLite + Twilio + Ollama - Multi-Site ERP

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General Industries Hub - Product Manual & User Catalogue

Version 2.0 - Multi-Site Warehouse Inventory ERP **Document Scope:** Complete operational reference for every role, page, tab, and element built into the system.

Table of Contents

1. [Introduction & System Overview](#1-introduction--system-overview)

2. [Roles, Permissions & Users]

1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a **multi-site warehouse inventory ERP** built on Streamlit + SQLite. It tracks every material movement (consumption, receipt, return, adjustment) through a two-stage approval ledger, enforcing real-time stock, FEFO (First-Expiry-First-Out) lot discipline, valuation, and audit trail across every site you manage.

1.2 Core principles

Principle	What it means
Identity math, not stored counters	$Current_Stock = Total_Received - Total_Consumed - Total_Returned$. The number is always computed from the immutable movement ledger - never written to a column.
Two-stage approval	Storekeeper is in charge of pending issues -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action writes to <code>system_audit_log</code> with username + timestamp + details. Even deletions leave a trace.
Self-healing schema	The DB layer automatically adds missing columns/tables on startup. You never need to run migrations manually.

1.3 The transaction lifecycle (the heart of the system)

```

????????????????????????????????????????????????????????????????????????????????????
? Store Keeper      ?????? pending_issues      ?????? HOD EOD Review      ?????? consumption      ?
? enters issue      ?   ? (status='pending_     ?   ? (edit/approve/       ?   ? (permanent      ?
?                   ?   ? hod')                  ?   ? reject/commit)       ?   ? ledger)         ?
????????????????????????????????????????????????????????????????????????????????????
                                     ?
                                     ?
                                     ?????????????????????????????????????????
                                     ? WhatsApp alerts ?
                                     ? + audit log      ?
                                     ?????????????????????????????????????????

```

The same shape applies to receipts (`pending_receipts` -> `receipts`) and adjustments (`stock_adjustments` -> `consumption/receipts`).

1.4 Currency, dates, units

- Currency: SAR (Saudi Riyal). All money is stored as REAL and displayed via `format_sar()`.

- Dates:

2. Roles, Permissions & Page Access

2.1 Role hierarchy

```
store_keeper (0) < supervisor (1) < hod (2) < admin (3)
```

Higher roles inherit all lower-role permissions plus their own. The hierarchy lives in `config.py:ROLE_HIERARCHY`.

2.2 Page access matrix

Page	Store Keeper	Supervisor	HOD	Admin
Live Dashboard	[X]	[OK]	[OK]	[OK]
Entry Log	[OK]	[OK]	[OK]	[OK]
HOD Portal	[X]	[X]	[OK]	[X] (hidden - Admin uses Admin Portal)
Admin Portal	[X]	[X]	[X]	[OK]
Reports	[X]	[OK]	[OK]	[OK]

2.3 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.
HOD	Their own site only - but they can REQUEST material from other sites (Cross-Site tab).
Admin	All sites globally - has the "All Sites" filter on every multi-site view.

2.4 Default seeded accounts

The first time the app starts, these accounts are created. **Change the passwords immediately.**

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Username text box - your assigned username (case-sensitive).

- Password text box

Registration form elements:

- Username - must be unique

- Password

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v2.0.0
Role card	Your username + role badge	Color-coded: green=Store Keeper, blue=Supervisor, purple=HOD, gold=Admin
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
render_brand_header	Live Dashboard, Entry Log	Gold subtitle accent + current date
render_brand_header_hod	HOD Portal	Purple subtitle accent + current date
render_brand_header_admin	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

- Type selectbox - Bug Report / Feature Request

- Page o
Admin

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] **OVERDUE ITEMS - Action Required:** N borrowed item(s) past expected return: <item names>.
Go to the **Returnable Items** tab to follow up.

4. Store Keeper Manual

The Store Keeper is the warehouse-floor operator. They see only the **Entry Log** page and the sidebar shell.

4.1 Pages visible

- Entry Log (only)

4.2 Entry Log - Tab structure

The Entry Log has **four tabs**:

1. Consumption Log - record material issued out

2. Receipts

4.3 Entry Log -> Consumption Log

This is where every material consumed by site operations is recorded.

4.3.1 Top section: Bulk OCR upload (expander)

Upload Handwritten Consumption List (OCR) - for when you have a handwritten list to bulk-stage instead of typing each row.

Element	Purpose
Input method radio	Switch between "Image upload (vision AI)" and "Paste text"
File uploader	Upload PNG/JPG/JPEG/WEBP of handwritten list (vision model reads it)
? Extract rows from image button	Triggers vision AI extraction -> preview rows
Paste text area	Alternative - paste tab/comma/pipe-separated rows
? Parse pasted rows button	Parse the textarea
OCR Review grid	Editable preview of extracted rows with ambiguous-match pickers
Confirm All & Stage button	Pushes all reviewed rows into pending_issues

4.3.2 Mobile camera barcode/QR scanner (expander)

Barcode / QR Scanner - opens the browser camera and reads SAP codes off labels.

Element	Purpose
Camera feed	Live preview (requires camera permission)
Manual SAP code entry text box	Fallback if camera unavailable - also accepts pasted scanner output
Detected code display	Shows the last successfully scanned code in green

When a code is scanned, the material selectbox auto-populates so you don't have to search.

4.3.3 ? Scan / Add New Item to Queue (main expander, expanded by default)

This is the workhorse panel. Reading top to bottom:

A. Recently-scanned ring buffer

Element	Purpose
?? Recent: pills	Up to 5 quick-tap buttons showing your last 5 items (SAP code + truncated description). Tapping one auto-fills the material selectbox.

B. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Type-ahead search across the inventory master. Format: [SAP_Code] Description.

When you pick a material, two cards appear automatically:

C. Item Snapshot card

A compact dark card showing:

- SAP code (gold monospace)

- Material

D. FEFO panel (lot suggestion)

If lots exist for this item at your site:

Element	Purpose
FEFO - First Expiry, First Out header (amber)	Banner
Per-lot row	Lot number (monospace) - xqty (dim) - "Exp <date>" (amber if <90d, red if expired) - USE FIRST badge on the first row

The system will automatically attach the top-FEFO lot number to your consumption row when you submit.

E. Pull from a different lot (FEFO override) - expander

Only renders when 2 or more open lots exist for this item at your site.

Element	Purpose
Lot to pull from instead selectbox	Default: "Keep FEFO suggestion". Other options: all other open lots with expiry + remaining qty shown
Reason for override text box (200 char max)	Mandatory. Min 5 characters to activate the override
Amber confirmation banner	Appears once override is active: "FEFO override active: pulling from <chosen> instead of <suggested>"

When activated:

- Your consumption row gets Lot_Number = chosen + FEFO_Override = reason

- Audit log

F. 2. Fill Entry Details (dynamic form)

Fields appear based on the [pending_issues](#) schema. Every required field is marked with *.

Field	Type	Purpose
Date	Date picker (default: today)	When the consumption happened
Quantity	Number input (min 0.1)	How much you're issuing. Above this field is a stock badge showing current site stock; a red [!] warning appears if
Work_Type	Selectbox (sourced from system_settings.Work_Type)	Classification: PM Work, Breakdown, Project, Shutdown, Inspection, etc.
Issued_By	Text (auto-filled with your username)	Who handed it over
Issued_To	Text (smart-defaults from your last entry)	Recipient personnel/department
Tank_No	Text	Equipment tag if applicable
Serial_No	Text (optional)	Item serial number if tracked
PR_Number	Text (smart-default)	Linked purchase request if applicable
Remarks	Text (optional)	Free-form notes

Hidden side-effects:

- Site_ID is auto-set to your site

- Lot_Nu

G. [!] Override Expiry Warning (amber card with checkbox)

If the system detects a short-dated batch at your site for this item, it will **hard-block** the Add to Grid button until you check this box, confirming you've physically pulled from the expiring batch first.

H. Add to Grid ?? button

Validates that:

- A material is selected

- No req

On success: inserts a `pending_issues` row with `status='draft'`, toasts [OK] Added to staging queue, reruns the page.

4.3.4 Staging Queue section

Element	Purpose
Staging Queue header + count badge	Shows how many draft rows you've accumulated
Data editor table	Edit any row inline before submitting
Save Draft Edits button	Persists edits in-place (rows stay as drafts)
Submit Grid to HOD button	Flips all draft rows to <code>status='pending_hod'</code> + queues a WhatsApp alert to the site HOD with the full item list

After submission, the rows are locked from your view - they appear in HOD Portal -> EOD Commit tab for approval.

4.4 Entry Log -> Receipt Staging

For when materials arrive at your site.

4.4.1 Upload Delivery Note (OCR) - expander

Identical lane structure to consumption OCR - upload an image of the delivery note OR paste text. Vision AI extracts rows for bulk staging.

4.4.2 ? Add Receipt to Queue - expander (expanded by default)**A. 1. Select Material**

Element	Purpose
Search by SAP Code or Description selectbox	Same type-ahead search as consumption

When picked, a blue-tinted info card shows:

- Mat Code

- UOM

B. 2. Fill Receipt Details (dynamic form)

Field	Type	Purpose
Date	Date picker (today)	Delivery date
Quantity	Number input	Units received
Supplier	Text (optional)	Vendor name
Lot_Number	Text (optional)	If blank + Expiry_Date is set, system auto-generates LOT-YYYYMMDD-SAP
Expiry_Date	Date picker (optional)	Lot expiry - when set, triggers lot master entry
PR_Number	Text (optional)	Link to an open Purchase Request
Serial_No / Vehicle_No / DN_No / Pallet_No / Mob_From / Mob_To / Prepared_by / Driver_Name	Text (optional)	Logistics tracking fields
Remarks	Text	Free notes

C. Add to Receipt Queue ?? button

Validates: material picked, required fields populated. Inserts a `pending_receipts` row with `status='draft'`.

4.4.3 Receipt Draft Queue section

Element	Purpose
Receipt Draft Queue header + count badge	Number of drafts you've accumulated
Data editor table	Inline edit before submit
Save Draft Edits button	Persists
Submit to HOD for Approval button	Flips drafts to <code>status='pending_hod'</code> + WhatsApp alert to HOD with item list

4.5 Entry Log -> Returnable Items

For tools, gauges, fittings temporarily handed to personnel - items that should come back.

4.5.1 ? Issue a Returnable Item - expander (expanded by default)

Field	Type	Purpose
Material / Tool Name	Text	What you're handing out
UOM	Text (e.g., Pcs, Set)	Unit
Quantity	Number (min 0.1)	How many
Borrower Name	Text	Person taking custody
Borrower WhatsApp Number	Text (optional, +966 ...)	For overdue alerts
Expected Return Date	Date (default: tomorrow)	When you expect it back
Expected Return Time	Selectbox: 04:15 PM, 06:15 PM,	

Custom Time...

Time-of-day expectation

Field	Type	Purpose
Issue Item button	Validates name+borrower, writes to returnable_items	

4.5.2 Currently Borrowed Items section

Shows all borrowed (not yet returned) items at your site.

- Overdue red banner - when one or more items are past their expected return time

- Borrowed Time, Location

4.6 Entry Log -> Stock Count

For reconciling physical shelf count with system stock when they don't match (damage, expiry disposal, miscount, found-extra).

4.6.1 1. Select Material to Count

Element	Purpose
Search by SAP Code or Description selectbox	Pick the material to reconcile
Help caption	Explains when to use this tab

4.6.2 2. Enter Count Details (only renders after picking a material)

Left column

Element	Purpose
System Qty card	Read-only dashboard showing current stock at your site (auto-fetched)
? Counted Qty input	What you actually count on the shelf right now (defaults to system qty)
Variance preview banner	Color-coded: green "? Found N extra" or red "? Short by N" or grey "No variance"

Right column

Element	Purpose
Reason Code selectbox	9 options. Default depends on variance direction.
Notes textarea (300 char max)	Free explanation: "found behind shelf 3", "damaged in transit", etc.

Reason codes:

Code	Label
cycle_count	Cycle count correction
damaged	? Damaged / unusable
expired_disposal	? Expired - disposed
miscount_in	? Miscount - found extra
miscount_out	? Miscount - short
lost	? Lost / unaccounted
theft	? Suspected theft
return_to_supplier	<-? Returned to supplier
other	? Other (see notes)

4.6.3 Submit section

Element	Purpose
Amber warning banner	"Submitting this sends the count to your HOD for approval. No stock changes until they approve."
Submit Count for HOD Approval button	Disabled while variance = 0. On success: writes to stock_adjustments with status='pending_hod', queues WhatsApp to site HOD with count details, audit-logs.

4.6.4 Recent Adjustments at This Site (history)

Read-only table showing: ID, SAP Code, Material Name, Variance, Reason, Status, Submitted By, Submitted At.

4.7 Store Keeper - Use Cases

Use Case 1: Issue 10 Pipe Gaskets to PM Work

1. Log in -> Entry Log -> Consumption Log tab

2. Open ?

The row(s) are now in HOD's EOD queue. Stock won't decrease until HOD commits EOD.

Use Case 2: Receive a delivery of 200 units with expiry date

1. Entry Log -> Receipt Staging tab

2. Open ?

Use Case 3: Found 5 extra units of an item (physical count > system)

1. Entry Log -> Stock Count tab

2. Pick th

HOD will approve in HOD Portal -> Adjustments. After approval, your live stock matches reality and a synthetic receipt row of +5 is posted to the ledger.

Use Case 4: Issue a tool to a borrower

1. Entry Log -> Returnable Items tab

2. ? Issue

If the item is not returned by 06:15 PM, the borrower (and you) get a WhatsApp alert.

Use Case 5: FEFO override (the right bin is blocked)

1. Entry Log -> Consumption Log -> pick material with 2+ open lots

2. Scroll t

Your HOD gets a real-time WhatsApp alert documenting the override.

4.8 Store Keeper - FAQ

Q: I can't see Live Dashboard, HOD Portal, or Reports. Did I lose access? A: No - Store Keepers only have access to Entry Log by design. If you need to view stock, ask your Supervisor or HOD.

Q: I submitted to HOD by mistake. Can I cancel? A: Not directly. Contact your HOD - they can reject the row in their EOD Commit review.

Q: The "Add to Grid" button is disabled or the form rejects my entry. A: Check three things: (a) is quantity > current stock? (b) is the expiry override needed but unchecked? (c) are all required * fields filled?

Q: I tried to issue 20 but the system says only 15 available. A: Believe the system. Either physically recount, or if there is genuinely 20 and the system disagrees, use the Stock Count tab to log a +5 adjustment.

Q: I don't have a barcode scanner. How do I quick-find items? A: Type a partial SAP code or description in the selectbox - it has type-ahead search. Or use the Recently-scanned pills if you've used the item recently.

Q: WhatsApp alerts aren't reaching the HOD when I submit. A: HOD's phone number must be set in their user profile. Tell the HOD to update it via Admin (or check with Admin directly).

Q: My borrowed item is overdue but I haven't received an alert. A: WhatsApp alerts are sent by a background worker (`whatsapp_worker.py`) - check with Admin that it's running.

Q: I uploaded an OCR image and it extracted wrong items. A: The OCR review grid lets you fix every row before staging. Pick the correct material from the suggested-matches dropdown, or type it manually.

Q: How do I see my recent activity? A: Submitted/approved items don't show on your screen. Ask HOD or Admin to filter the audit log by your username.

5. Supervisor Manual

The Supervisor monitors a single site's stock, generates reports, and provides oversight. They cannot approve transactions (that's HOD).

5.1 Pages visible

- Live Dashboard

- Entry

5.2 Live Dashboard

The Live Dashboard is the at-a-glance view of every catalogue item with current stock, value, and status.

5.2.1 Brand header

`render_brand_header("Live Warehouse Stock Dashboard")` - gold subtitle + today's date.

5.2.2 Hero metric strip (4 cards)

Card	Source	Tone logic
Catalogue items	Count of rows in inventory	Neutral
Total stock value	<code>cached_total_inventory_value()</code> (Sum of <code>Current_Stock</code> × <code>Unit_Cost</code> , all sites) - formatted as SAR 1,234 / SAR	Neutral. Delta: "standard cost - all sites" or "set Unit_Cost in Admin -> DB Editor" if 0
Below minimum	Count from <code>cached_low_stock_items()</code>	Green=0, Amber<10, Red>=10. Delta: "all healthy" or "needs reorder"
Expiring / expired	Count from <code>cached_short_dated_stock()</code>	Green=0, Amber<10, Red>=10. Delta: "shelf-life clear" or "review HOD Portal"

5.2.3 Ask in plain English (AI search) - expander (if `AI_ENABLED`)

If Ollama is running locally with the `qwen2.5-coder:7b` model:

Element	Purpose
Your question text input	Natural language query, e.g., "items below minimum at site B"
Search button	Translates to SQL via local LLM, runs read-only
Clear button	Resets the panel
Result table	Returned rows
"Show SQL the AI generated" expander	Shows the safe SELECT the LLM produced

If Ollama is NOT running: an amber card displays setup instructions (`ollama serve`).

5.2.4 Burn alert banner

Appears at the top of the table area when there are items burning to zero within the configured window (default 7 days). Color-coded amber/red.

5.2.5 Main inventory grid (AgGrid)

Column	Source	Notes
SAP_Code	inventory.SAP_Code	Primary key
Equipment_Description	inventory.Equipment_Description	Material name
UOM	inventory.UOM	Unit of measure
Total_Returned	Computed	Sum from returns table
Current_Stock	Computed identity math	Received - Consumed - Returned
Minimum_Qty	inventory.Minimum_Qty	Reorder threshold
Unit_Cost	inventory.Unit_Cost	Set via Admin -> DB Editor
Stock_Value	Computed	Current_Stock × Unit_Cost, rounded 2 dp
Status	Computed badge	OK / Low / Below Min / Empty with colored pill via STATUS_BADGE_JS

You can sort by any column - sorting by `Stock_Value` shows your biggest SAR exposure first.

5.2.6 Expanders below the grid

Expander	Content
Stock vs Minimum Threshold	Horizontal bar chart, sorted by criticality. Each bar = item; gold line markers = minimum threshold. Color: red=empty, amber=below min, orange=close, green=ok. Capped at 20
Burn Rate Forecast (30-Day)	Bar chart with daily-burn-rate bars + vertical line at the 30-day alert threshold. Colors: red <10 days, amber <30, green >30.
Top Consumed Items	Bar chart of top 10 items by 30-day consumption, blue->gold gradient.

5.3 Reports

Supervisor sees the full Reports module but with site scope locked to their site (cannot pick "All Sites"). See §8 for the full Reports reference.

5.4 Supervisor - Use Cases

Use Case 1: Daily morning check

1. Log in -> Live Dashboard

2. Read t

Use Case 2: Generate end-of-month report for management

1. Reports -> Generate Report

2. Pick ?

Use Case 3: Investigate why an item ran out

1. Live Dashboard -> search the item (or click its row)

2. Open

5.5 Supervisor - FAQ

Q: I can only see my own site. Why can't I see other sites' stock? A: By design - site isolation. Only Admin sees all sites. If you need to know another site's stock for a transfer, ask your HOD to file a Cross-Site Request.

Q: Total stock value shows SAR 0. What's wrong? A: Items have no `Unit_Cost` set. Ask Admin to enter costs in Admin -> Master DB Editor -> inventory table.

Q: AI search says "Ollama not running." A: Local AI is optional. Admin must install Ollama and pull the `qwen2.5-coder:7b` model. Without it, the system still works - you just type into the standard grid filters.

Q: The Stock_Value column is empty for some items. A: Those items have `Unit_Cost = 0`. The valuation is correct - they're tracked in qty but not in money.

Q: I can't approve any transactions. A: Correct. Supervisors monitor; HODs approve. Talk to your HOD.

Q: Can I export the Live Dashboard grid? A: Yes - AgGrid has a built-in CSV export (right-click the grid). For a formal report, use Reports -> Generate.

6. HOD (Head of Department) Manual

The HOD owns their site's inventory ledger. Every transaction flows through their approval. The HOD Portal has **10 tabs**.

6.1 Pages visible

- Live Dashboard (see §5.2)

- Entry

6.2 HOD Portal overview

6.2.1 Page header

`render_brand_header_hod("HOD Management Portal")` - purple subtitle + today's date.

- Page title: "HOD Portal" + "Managing Site: <SITE_ID>"

6.2.2 Hero metric strip (4 cards)

Card	Source	Notes
Site stock value	<code>cached_total_inventory_value(site_id)</code>	Delta: <SAR> consumed (30d) - site-scoped
Below minimum (site)	<code>cached_low_stock_items(site_id)</code> count	Green/amber/red
Expiring / expired	<code>cached_short_dated_stock(site_id)</code> count	Green/amber/red
Pending receipts to approve	Count from <code>pending_receipts</code> where <code>status='pending_hod'</code> at your site	Neutral or amber

6.2.3 Tab strip (10 tabs)

1. EOD Commit

2. Cross

6.3 HOD Portal -> EOD Commit

The single most consequential action in the system: committing the day's staged consumption to the permanent ledger.

6.3.1 Top section

Element	Purpose
Header	" End-of-Day Commit (<SITE>)" with intro caption

6.3.2 4-card stat strip

Card	Source
Total entries	Total pending rows at your site
? Pending	Status = pending
[!] Flagged	Status = flagged
[OK] Approved	Status = approved

6.3.3 Filter pills

Pill	Action
All / Pending / Flagged / Approved / Rejected	Filters the table below to that status

6.3.4 Top action bar

Button	Action
[OK] Approve All Pending	Sets every pending row to status='approved'. Disabled when count = 0.
Commit EOD to Master	Opens the type-COMMIT modal. See §6.3.7

6.3.5 Main pending-issues table

Custom dark-themed HTML table showing **every column from the staging row** (joined with inventory for descriptions):

Column	Source
Date	pending_issues.Date
SAP	pending_issues.SAP_Code
Mat Code	inventory.Material_Code
Material	inventory.Equipment_Description
UOM	inventory.UOM
Qty	pending_issues.Quantity
Work Type	pending_issues.Work_Type

Column	Source
PR	pending_issues.PR_Number
Tank	pending_issues.Tank_No
Serial	pending_issues.Serial_No
Issued By	pending_issues.Issued_By
Issued To	pending_issues.Issued_To
Remarks	pending_issues.Remarks
Status	Colored pill (pending / flagged / approved / rejected / committed)

The table also reflects `Lot_Number` and `FEFO_Override` columns when present (visible by editing the row).

6.3.6 Row Actions panel (per-row Approve/Reject)

Below the table, top 20 actionable rows render as cards:

Element	Purpose
Card line 1	SAP - Material Code - Material name - Qty UOM
Card line 2	Date - Work Type - PR - Tank - Serial - By - To - Remarks (only fields with values)
[OK] button	Approve this single row (sets status='approved')
[X] button	Reject this single row (sets status='rejected')

6.3.7 EOD Commit confirmation modal (type-COMMIT dialog)

When you click **Commit EOD to Master**:

Step 1 - pre-flight: negative-stock guard

The system calls `validate_eod_no_negative_stock()`. If any items would push stock below zero:

Element	Purpose
? Red error banner	"Cannot commit - N item(s) would go negative"
Violation table	SAP - Material - Current Stock - Trying to Consume - Deficit
? Hint banner	"Common fixes: receive inbound stock first, raise a stock adjustment, or reduce the consumption to fit available stock."
Close button	Cancels - fix the staging rows and try again

Step 2 - confirm (only if no violations)

Element	Purpose
Warning banner	"You are about to commit N pending row(s)..."
Type COMMIT text input	Must type exactly COMMIT to enable the button
Cancel button	Abort
Confirm Commit button (red)	Calls <code>commit_eod()</code> - moves rows to consumption, deletes from <code>pending_issues</code> , busts caches, queues a post-EOD

low-stock alert to the HOD if applicable.

After commit: ? balloons animation + success toast.

6.3.8 Flagged-items banner

If there are flagged rows: amber banner appears below the table reminding the HOD to verify with the store keeper before committing.

6.4 HOD Portal -> Cross-Site

For requesting material from another branch.

6.4.1 Top section

Element	Purpose
Intro caption	Explains the > 5-item escalation rule

6.4.2 Single-target inquiry flow (left/right columns)

Left column:

Element	Purpose
Select Target Branch selectbox	List of all sites except yours
Select Material selectbox	Type-ahead search across inventory
Quantity Needed number input	Min 1
Justification / Notes textarea	Why you need this transfer

Right column:

Element	Purpose
Live Stock at <target> header	Shows availability at the target site
Available Quantity metric	Live stock at target
Suggested Transfer Qty metric	min(requested, available)
? Add to List button	Adds to the cart

6.4.3 ? Your Request Cart section

Element	Purpose
Cart table (styled HTML)	Columns: Target Site - SAP - Description - Qty - Notes
Submit All Requests to Admin button	Creates a requests row per cart item, queues WhatsApp to all admins, AND (if >5 items) escalates to the target-site HOD via separate WhatsApp + audit log
? Clear List button	Empties the cart

6.4.4 Incoming Cross-Site Requests panel (bottom)

Where a HOD sees requests **addressed to their site** from other HODs.

Element	Purpose
Intro caption	"Requests addressed to <site>"
Incoming table (styled HTML)	From - SAP - Material - Qty - By - Notes - When
Per-row Approve / Reject controls (top 10)	[OK] Approve and [X] Reject buttons
Approval action	Calls <code>update_request_status(status='approved')</code> + audit
Rejection action	Same with <code>status='rejected'</code>

6.5 HOD Portal -> Burn Rate

Predictive analysis of which items will run out and when.

6.5.1 Compact horizontal-bar chart (top 10)

A sleek card with:

- Header row: "Monthly Consumption - Top 10" + legend chips (red=High / amber=Mid / green=Low)

- Per-row
max) -

6.5.2 Burn alert banner (above the Plotly chart)

Auto-renders if items are projected to hit zero within the alert window.

6.5.3 Full Plotly burn rate chart

Daily-burn-rate bars with vertical line at 30-day alert threshold. Color-coded.

6.5.4 Detailed Forecast Table (AgGrid)

Columns: `SAPCode` - `EquipmentDescription` - `UOM` - `CurrentStock` - `DailyBurnRate` - `DaysRemaining`

Sortable + exportable.

6.6 HOD Portal -> ? Pending Receipts

Approve or reject receipts staged by Store Keepers.

6.6.1 Header + count banner

Amber banner: "? N pending receipt(s) require your approval before stock levels update."

6.6.2 [OK] Approve All button

Bulk: calls `commit_pending_receipts(site_id, username)` which:

1. Iterates every pending receipt

6.6.3 Receipt table

Columns: Date - SAP - EquipmentDescription - UOM - Quantity - Supplier - PRNumber - Site_ID - **Status pill**

6.6.4 Per-row Reject controls (top 10)

Element	Purpose
Card line	SAP - Qty UOM - Supplier
[X] Reject button	Soft-reject (row stays as status='rejected'), preserves audit, audit-logs with reason

6.7 HOD Portal -> Adjustments

Approve physical-count reconciliations submitted by Store Keepers from the Entry Log -> Stock Count tab.

6.7.1 3-card stat strip

Card	Source
? Pending	Count of pending adjustments at your site
? Shortfalls	Variance < 0 count
? Surpluses	Variance > 0 count

6.7.2 Pending adjustments table

Columns: # - SAP - Material - UOM - System - Counted - **Variance** (red for negative, green for positive) - Reason - By - When

6.7.3 Per-row Approve/Reject

Below the table:

Element	Purpose
Card line 1	SAP - Material - System X -> Counted Y - Variance
Card line 2	Reason label - submitted by - notes
[OK] button	Calls approve_stock_adjustment(adj_id, approver). Atomically posts a synthetic ledger row - if variance < 0, inserts into consumption with Work_Type=STOCK_ADJUSTMENT; if > 0, inserts into

receipts with Supplier=STOCK_ADJUSTMENT. The posted_txn_ref ('C:rowid' or 'R:rowid') links back to the adjustment for full audit.

Element	Purpose
[X] button	Soft-reject. Row stays as status='rejected' for audit.

6.7.4 Recent Adjustments History (last 30)

Columns: # - SAP - Material - Variance - Reason - **Status pill** - Submitted By - Approved By - Ledger Ref

6.8 HOD Portal -> Purchase Requests

Manage Purchase Requests (PRs) for your site.

6.8.1 ? Create New PR (manual entry) - expander

Field	Purpose
PR Number	Free text, required (e.g., 3001234567)
Material selectbox	Picks from inventory; auto-fills Material_Code, Material_Name, UOM via blue-tinted info card
Requested Qty	Min 0.01
Preferred Supplier	Optional
Estimated Cost (SAR)	Optional
UOM	Auto-filled, editable
Notes	Optional
Create PR Draft button	Calls insert_manual_pr(...). Sets status='open', workflow_state='draft'

6.8.2 ? Upload PR PDF (auto-extract via pdfplumber) - expander

Element	Purpose
File uploader (PDF only)	Upload Purchase Request PDF
Process Upload button	Calls process_pr_pdf() - extracts PR number + Material Codes (GI-XXXXXXX pattern), matches to inventory's Material_Code, inserts pr_master rows
Result banner	Green success or amber warning (if some materials unmatched)

6.8.3 Current PRs table

Columns: PR No. - SAP - Material - UOM - Qty Req - Qty Pending - Supplier - Est. SAR - **Workflow pill**

Pending_Qty = Requested_Qty ? SUM(receipts where PR_Number matches)

6.8.4 ? Notify Logistics section

When at least one PR has status='open':

Element	Purpose
Select PR for Actions selectbox	Pick from open PRs
? Draft Outlook Email button	Opens email client with HTML table of pending items. Mac: opens Mail.app via AppleScript with formatted HTML table. Windows: opens Outlook via COM with HTMLBody.
Download PR PDF button	Generates a PDF record with columns: Material Code - Description - Req. Qty - Received - Pending - Status. Auto-saved metadata: Site, Generated By, Timestamp.

6.9 HOD Portal -> Receive Material

For HOD-direct receipts (when a HOD logs a delivery themselves, e.g., direct purchases or HQ deliveries).

6.9.1 PR linker

Element	Purpose
? Link to Open PR selectbox	Filters the material list to items in that PR, or "None (Direct Purchase)" for free-form

6.9.2 Receipt form (st.form, clears on submit)

Field	Type	Purpose
Select Material	Selectbox	From inventory or filtered to PR items
Quantity Received	Number (min 0.1)	Units arrived
Delivery Date	Date (today)	When
Expiry Date	Date (optional)	Lot expiry trigger
Logistics extras	Text inputs	All non-system columns on receipts (Vehicle_No, Driver_Name, DN_No, Supplier, Remarks, etc.)
Save Receipt button	Calls process_receipt_delivery() directly (no staging) - instantly posts to receipts AND creates a lots master row if Expiry_Date is set. Auto-closes	

6.9.3 Receipt History (last 50) PR if fulfilled. Busts caches.

Below the form, an AgGrid showing recent receipts for this site: rowid - Date - SAP - Material - Quantity - Supplier - PRNumber - ExpiryDate.

6.10 HOD Portal -> [!] Shelf-Life

Lots at risk of expiry.

6.10.1 3-card stat strip

Card	Source
Expired	Lots with days_left < 0
? Critical (?30d)	Days_left 0-30
Warning (?90d)	Days_left 31-90

6.10.2 Action-required banner

When `expired_n > 0`: red banner instructing physical isolation + disposal.

6.10.3 Shelf-life table (top 30, sorted by days-left ascending)

Columns: SAP - Material - Lot - Qty - Expiry (color-coded) - **Days Left** - **Status pill**

6.10.4 ? Log Disposal button

Bulk-logs an audit event for the expired count. Note: this does NOT post a stock adjustment automatically - for proper inventory reduction, use the Adjustments flow with reason `expired_disposal`.

6.11 HOD Portal -> Notifications

Manual WhatsApp sends + alert threshold tuning.

6.11.1 Send Manual WhatsApp card

Field	Purpose
Recipient phone number	E.g., +966 5X XXX XXXX
Message textarea	Up to several hundred characters
Send WhatsApp button	Queues to <code>whatsapp_queue</code> with <code>status='pending'</code> ; the background worker picks it up. Audit-logs <code>MANUAL_WHATSAPP</code> .

6.11.2 Alert Thresholds card

3 sliders writing to `app_settings`:

Slider	Range	Default	Stored key
Low stock alert (days of supply)	1-60	5	<code>low_stock_days</code>
Burn-rate warning (days remaining)	1-60	7	<code>burn_alert_days</code>
Expiry warning (days before)	1-120	30	<code>expiry_warn_days</code>

Save Thresholds button - persists + audit-logs `UPDATE_THRESHOLDS`.

> Note: The Notification Log table moved to Admin Portal -> WhatsApp Console for global visibility.

6.12 HOD Portal -> [OK] My Requests

Outbound cross-site requests YOU have made.

6.12.1 Outbound requests table (styled HTML)

Columns: # - To Site - SAP - Qty - **Status pill** - Created

6.12.2 Mark Incoming Transfers as Received

When at least one request has status='approved':

Element	Purpose
Select Approved Request selectbox	List of approved request IDs
Confirm Delivery Received button	Calls update_request_status(status='fulfilled') + busts caches. This is where you confirm the transfer physically arrived.

6.13 HOD - Use Cases

Use Case 1: Approve and commit the day's consumption

1. HOD Portal -> EOD Commit

2. Review

Use Case 2: Approve a Store Keeper's physical count adjustment

1. HOD Portal -> Adjustments

2. Read t

Use Case 3: Request material from another site

1. HOD Portal -> Cross-Site

2. Pick ta

Use Case 4: Approve an incoming bulk request from another HOD

1. HOD Portal -> Cross-Site -> scroll to Incoming Cross-Site Requests

2. Review

Use Case 5: Raise a Purchase Request

Option A: manual entry

1. HOD Portal -> Purchase Requests

2. ? Crea

Option B: PDF upload

1. ? Upload PR PDF

2. Drop th

Use Case 6: Notify logistics about a pending PR

1. HOD Portal -> Purchase Requests

2. Select

6.14 HOD - FAQ

Q: The EOD commit modal won't let me commit - keeps showing the violation table. A: Pre-flight is blocking because consuming the staged amount would create negative stock. Either: (a) reduce qty on the violating rows in the EOD table, (b) commit pending receipts first to top up stock, (c) raise a stock adjustment to correct any system-vs-physical discrepancy.

Q: A Store Keeper claims they submitted but I don't see anything. A: Check the filter pills (you may

be filtering by Approved/Rejected only). Switch to "All" or "Pending".

Q: I approved an adjustment but the live stock didn't change. A: It should change within 30 seconds (cache TTL). Refresh the page. If it still doesn't, check audit log for the APPROVE_ADJUSTMENT entry to confirm the synthetic ledger row was posted.

Q: I got a WhatsApp about a "bulk cross-site request" - what is it? A: Another HOD at a different site has requested more than 5 items from your site. Go to Cross-Site -> Incoming to review.

Q: My burn rate forecast looks wrong - items aren't appearing. A: Items with fewer than ~3 consumption events in the last 30 days have insufficient data and may be omitted. Increase the window via Settings if needed.

Q: I can't see a receipt I just approved. Did it work? A: Approving via "Approve All" commits all pending receipts at once. Check the Live Dashboard for updated stock. The receipt now lives in `receipts` table, not `pending_receipts`.

Q: The Outlook email button does nothing on my Mac. A: Make sure Mail.app or Outlook is installed and the system has permission to open it. The fallback is `mailto:` (plain-text monospace table) which works in any browser default mail handler.

Q: PR PDF upload didn't recognize some items. A: The extractor matches material codes in the format `GI-XXXXXXX`. Items not in your inventory's `Material_Code` field will be flagged in the warning. Add the missing items to inventory first.

Q: Cross-site request stuck in 'pending' - why isn't Admin approving? A: Admin sees all pending in Admin Portal -> Pending Requests. If urgent, queue a WhatsApp from Notifications. Or for >5 items, the target-site HOD can also approve directly.

Q: I want to see all FEFO overrides logged by my Store Keepers. A: Filter Admin -> ? Audit Logs by action `FEFO_OVERRIDE`. Or query the `consumption` table for `FEFO_Override IS NOT NULL`.

7. Admin Manual

The Admin is the system owner. The Admin Portal has **9 tabs** covering operations, users, data, audit, integrations, and security. Admins do NOT see HOD Portal (intentional - Admin uses Admin Portal for cross-site work).

7.1 Pages visible

- Live Dashboard (global - sees all sites combined)

- Entry

7.2 Admin Portal overview

7.2.1 Page header

`render_brand_header_admin(...)` - gold subtitle + pulse status chip (green "All systems operational" or amber "Degraded - see Overview" if global low-stock count > 20).

7.2.2 Hero strip (3 cards)

Card	Source
Sites managed	Count of distinct sites
Pending cross-site requests	Count from requests where status='pending'
Critical items (all sites)	low + expiry count summed globally

7.2.3 Tab strip (9 tabs)

1. ?? Overview

2. Pending

7.3 Admin Portal -> ?? Overview

System health at a glance.

7.3.1 4-card KPI strip (technical)

Card	Source
?? DB size	File size of gi_database.db on disk (MB)
? Users	Count of non-suspended users
Total transactions	consumption + receipts row count
? Audit events	All-time count from system_audit_log

7.3.2 4-card valuation strip (financial)

Card	Source
Total stock value	cached_total_inventory_value() (all sites)
? Biggest-value site	cached_value_by_site().iloc[0] + share of total
30-day consumption value	cached_consumption_value(days=30)
Pending receipts value	Placeholder showing source: pr_master.Est_Cost_SAR

7.3.3 Service Health card

Per-service row with a pulse dot + status + note:

Service	Up indicator	Source
SQLite Database	Always up if reached	WAL mode - DB size MB
WhatsApp Queue	If whatsapp_queue reachable	Pending count or "queue clear"
Ollama / AI	OLLAMA_AVAILABLE if	"ready" / "not reachable"
Mail / SMTP	Mail Enabled	"Outlook + mailto fallback"

7.3.4 Database Stats card

8 row counts:

- Inventory items

- Consumed

7.3.5 Live Activity Feed (last 12)

Per-row card from system_audit_log:

- // severity icon (derived from action_type via _severity_from_action())

- Timestamp

7.4 Admin Portal -> Pending Requests

Approve or reject cross-site material requests from HODs.

7.4.1 Pending requests data editor

Editable table with a ?? `Select` checkbox column (all other columns disabled). Source: `get_pending_requests(status='pending')`.

7.4.2 Admin Notes text input

Optional for approve - **required for reject**. Free text passed to `update_request_status()`.

7.4.3 Action buttons

Button	Action
[OK] Approve Selected	For each selected row: looks up Material Code/Description, updates <code>requests.status='approved'</code> . Then for each unique requester: queues a WhatsApp with same site id and loopbu sets each status to 'approved'.
[X] Reject Selected	Same as above but sets each status to 'rejected'. Sites are queued a WhatsApp with 'Rejected' and a WhatsApp with 'Approved'.

7.5 Admin Portal -> ? Global Sites

Cross-site inventory viewer (read-only).

Element	Purpose
Site selectbox	"All Sites (Global)" or pick one
Inventory dataframe	Calls <code>cached_live_inventory(site_id=...)</code>

Useful before approving a cross-site request to confirm the target site really has the stock.

7.6 Admin Portal -> ? Users

Delegates entirely to `auth.render_user_management_tab()` which provides:

Section	Purpose
Pending registration requests	Approve / reject self-registered users
Active users table	All users with role, site, phone
Edit user form	Change role, site, phone, suspend/activate
Add user form	Create user directly (Admin shortcut)
Password reset action	Generates temporary password, audit-logged

7.7 Admin Portal -> ?? Master DB Editor

This is the most powerful tab. Use with care. Lets Admin view, edit, and modify any table's data and schema.

7.7.1 Table selector

Element	Purpose
Select Table dropdown	Lists every user table in the SQLite DB. Includes new ones: lots, stock_adjustments, bug_reports, app_settings, etc.

7.7.2 Action radio

3 modes:

Mode A: View / Edit Data

Element	Purpose
Row count caption	"<N> rows in <table>"
? Export as PDF button	Generates generate_universal_pdf() of the whole table
Data editor table	Edit any cell. Password hashes are masked as *****,
Print Label checkbox column	Only on inventory table - select items to print QR labels
Save Table Updates button	DELETE the table, INSERT_ALL from the edited dataframe. Audit-logs DB_EDIT. Busts inventory + settings caches.
?? Generate QR Labels for Selected button	Only on inventory - generate_qr_labels_pdf() for items where label checkbox is checked
Download QR Labels PDF button	Appears after generation

> [!] **The Save action is destructive.** It deletes all rows and re-inserts. If anything fails mid-write you may lose data. Industry best-practice is to make ledger tables immutable; this tab bypasses that - Admin discretion required.

Mode B: ? Add New Entry

Form generation depends on the table:

- For users: warns to use User Management tab instead (safer)

- For rec
Supplie

Mode C: Manage Columns

Element	Purpose
? Add Column section	Column name text input + Add button -> ALTER TABLE ADD COLUMN <name> TEXT
?? Rename Column section	Column dropdown + new name + Rename button -> ALTER TABLE RENAME COLUMN
? Drop Column section	Column dropdown + Delete button -> ALTER TABLE DROP COLUMN (SQLite supports this only on recent versions)

> The `users` table is protected - schema changes are blocked here ("managed by auth.py").

7.8 Admin Portal -> ? Audit Logs

Complete forensic record.

7.8.1 Filter row

Filter	Source
User selectbox	DISTINCT usernames + "All Users"
Action selectbox	DISTINCT action_types + "All Actions"
Target selectbox	DISTINCT target_tables + "All Targets"
Limit selectbox	50 / 100 / 500 / 1000

7.8.2 Search text input

Substring match across details + username + action_type.

7.8.3 Audit table (styled HTML)

Columns: // severity - Timestamp - User - **Action** (color-coded by severity) - Target pill - Detail (truncated, tooltiped)

Severity heuristic (from action_type):

- Critical (): any action containing FAIL, REJECT, DELETE, PURGE, EMERG, ROLLBACK, RESET, DESTRUCTIVE

- Warning

Expired-row red tint on critical events.

7.9 Admin Portal -> WhatsApp Console

Outbound message queue + manual sends + thresholds + event mapping.

7.9.1 Queue stats strip (4 cards)

Card	Source
[OK] Sent	whatsapp_queue where status='sent'
? Pending	status='pending'
Processing	status='processing'
[X] Failed	status='failed'

7.9.2 Send Manual WhatsApp card

Field	Purpose
Recipient phone	E.g., +966 5X XXX XXXX
Message textarea	The message to send
Send WhatsApp button	Queues to whatsapp_queue + audit-logs MANUAL_WHATSAPP

7.9.3 Alert Thresholds card (global)

Same 3 sliders as HOD Notifications (lowstockdays / burnalertdays / expirywarndays). Persists to `app_settings`. Admin's value is the global default.

7.9.4 Event -> Recipient (current wiring) - read-only summary

A reference table showing **what auto-triggers exist in the codebase**:

Event	Recipient	Role
Issue staging submitted	Site HOD	hod
Pending receipt submitted	Site HOD	hod
EOD committed	Site HOD	hod
Cross-site request created	All admins	admin
Cross-site bulk (>5 items)	Target site HOD	hod
Cross-site request approved	Requesting HOD	hod
Cross-site request rejected	Requesting HOD	hod
Returnable item overdue	Store Keeper	store_keeper
New access request	All admins	admin
Access request approved	Requesting user	store_keeper
Post-EOD low stock alert	Site HOD	hod

7.9.5 Outbound Queue Log (last 80)

Styled HTML table from `get_whatsapp_log(limit=80)`: **Status pill** - Recipient (monospace) - Message (truncated with tooltip) - Queued at - Sent at

7.10 Admin Portal -> Settings

7.10.1 Dropdown Manager - Work Types (expander, expanded by default)

Element	Purpose
Current Work Types caption	Comma-separated list from system_settings
New Work Type Name text input	Add a new option
Add to Dropdown button	INSERT into system_settings (category='Work_Type') + busts settings cache

7.10.2 Maintenance Mode card

Element	Purpose
Enable maintenance mode toggle	Persists to app_settings.maintenance_mode='1'
Status caption	"ACTIVE - Non-admin sessions will be told to come back later" or "Off"

Audit-logs TOGGLE_MAINTENANCE. (Enforcement at login is in `auth.py` and `main.py`.)

7.10.3 ?? Database Backup card

Element	Purpose
Last manual backup caption	From app_settings.last_backup_at
Backup Now button	Calls <code>shutil.copy2()</code> of <code>gi_database.db</code> into <code>backups/gi_database_YYYYMMDD_HHMMSS.db</code> . Updates <code>last_backup_at</code> + audit DB_BACKUP.

7.10.4 ? Site Management

Element	Purpose
Sites table (styled HTML)	Site Name - Code (first 4 chars upper) - Users count - Status (always "Active")
? Add New Site expander	Type name + button -> INSERT into system_settings (category='Site') + audit ADD_SITE

7.10.5 [!] Danger Zone (red bordered card)

Element	Purpose
Purge old draft pending_issues card	"Delete all pending_issues rows older than 30 days that are still status='draft'"
Type PURGE to confirm	Text input must equal PURGE to enable
Run Purge button	Executes the DELETE, audit-logs PURGE_DRAFTS

7.11 Admin Portal -> ? Access Control

7.11.1 ?? Recent Sign-Ins (last 10)

Per-row card from `system_audit_log` filtered to LOGIN/LOGINSUCCESS/LOG/INFAILED/LOGOUT:

- Pulse dot (green=success, red=failed)

- Userna

7.11.2 ? Force Password Reset card

Field	Purpose
Target user selectbox	From users table
New password	Type input (password mask)
Confirm	Must match
Amber warning	"User must log in again immediately."
? Reset Password button	Validates length ? 8 + match -> hash_password(new_pwd) -> UPDATE users SET password_hash + audit FORCE_PASSWORD_RESET

7.11.3 Security Policy

Read-only 2-column grid showing:

- Auth backend (bcrypt cost=12)

- Session

7.12 Admin - Use Cases

Use Case 1: Set initial Unit_Costs for valuation reports

1. Admin Portal -> ?? Master DB Editor

2. Select

Use Case 2: Approve a self-registered new user

1. Admin Portal -> ? Users

2. Pending

Use Case 3: Backup the database before a risky operation

1. Admin Portal -> Settings

2. Backu

Use Case 4: Investigate a suspicious series of FEFO overrides

1. Admin Portal -> ? Audit Logs

2. Action

Use Case 5: Configure WhatsApp alert thresholds

1. Admin Portal -> WhatsApp Console

2. Alert T

Use Case 6: Approve a batch of cross-site transfers

1. Admin Portal -> Pending Requests

2. Review

Use Case 7: Drop a deprecated column from a custom table

1. Admin Portal -> ?? Master DB Editor

2. Select

Use Case 8: Generate QR labels for new inventory items

1. Master DB Editor -> inventory

2. View /

7.13 Admin - FAQ

Q: How do I add a new site? A: Admin Portal -> Settings -> ? Site Management -> ? Add New Site. New sites are immediately available for user assignment.

Q: How do I rotate a user's password without their input? A: Admin Portal -> ? Access Control -> ? Force Password Reset. Pick user, type new password twice, click reset. They'll need to log in again with the new credentials.

Q: Audit log filter has too many users - how do I find a specific user's activity? A: Use the search text input at the bottom of the audit filter row - searches across user + action + details.

Q: WhatsApp queue is stuck - messages aren't being sent. A: The background worker ([whatsapp_worker.py](#)) must be running separately (not via Streamlit). Check via terminal: `python whatsapp_worker.py`. The queue table will fill but won't drain otherwise.

Q: My DB Editor save crashed mid-way and lost rows. A: Restore from [backups/](#). The Save action does DELETE-then-INSERT and is not crash-safe. Always backup before bulk edits.

Q: Maintenance Mode is on but non-admins can still log in. A: Enforcement requires a check in [auth.py](#). The toggle persists; ensure the auth gate respects [app_settings.maintenance_mode](#).

Q: Can I delete an audit log entry? A: No - they're meant to be immutable. If you need to free space, do a SQL purge with a date filter via DB Editor -> Manage Columns... but this is itself audited.

Q: A cross-site request is stuck pending and the requester is asking why. A: Admin Portal -> Pending Requests. Confirm it's actually in the queue. If yes, approve or reject with notes. The HOD will get a WhatsApp.

Q: I want to set per-user 2FA. A: Currently unsupported in the security model (placeholder shown in Access Control). Future enhancement.

Q: How do I know if Ollama (local AI) is up? A: Admin Portal -> ?? Overview -> Service Health card. If "Ollama / AI" pulses green, it's reachable.

Q: What's the difference between approving a Pending Receipt vs Pending Issue? A: Receipts ADD to stock (your team RECEIVED material). Issues SUBTRACT from stock (your team CONSUMED material). Both flow through approval. Issues commit via EOD (batch); Receipts commit individually via Pending Receipts tab.

Q: Stock_Value column shows nothing on the dashboard. A: Inventory items have `Unit_Cost = 0`. Use DB Editor to set costs.

Q: A HOD set thresholds different from mine - whose wins? A: Both HOD and Admin write to the same [app_settings](#) keys. Last-writer-wins. Decide a policy and lock it via convention.

Q: How big can the database grow before performance degrades? A: SQLite with WAL comfortably handles 10-25 concurrent users. The codebase is structured so [database.py](#) is the only SQL surface - migrating to Postgres later is a contained change.

8. Reports Module - Detailed Reference

Available to: **Supervisor, HOD, Admin**. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

`render_brand_header("Reports & Analytics"). Tabs:`

1. Generate Report

2. ? Sche

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	Description	Source
Daily Consumption	All material issues in the window, grouped by work type	<code>report_daily_consumption()</code>
? Monthly Summary	Per-SAP opening, issued, received, closing stock + SAR value columns (Issued, Received, Closing)	<code>report_monthly_summary()</code>
[!] Low Stock Alert	Materials below minimum with shortfall	<code>get_low_stock_items()</code>
Burn Rate Analysis	30-day trends, days-of-supply per item	<code>cached_burn_rate_and_forecast()</code>
Inventory Valuation	Per-item Stock_Value sorted descending; summary with top-10 share	<code>get_inventory_valuation()</code>
Shelf-Life / Expiry	Lots by expiry -	

expired/critical/warning buckets

get_short_dated_stock()

Report	Description	Source
PR Status Report	All PRs with status, workflow_state, Est_Cost_SAR	report_pr_status()
[OK] FEFO Compliance	Audit of FEFO adherence - picks vs oldest lot	report_fefo_compliance()
? Full Audit Report	Complete system_audit_log for the date range	report_audit_export()

8.2.3 ? Generate Report button

Calls `_run_report(type, from, to, site_id)` -> returns (DataFrame, summary_dict). Loading state shows shimmer.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	"<Type> - <site>" + "<from> -> <to>"
v Download <fmt> button	Encodes via generate_report_pdf/excel/csv and triggers st.download_button
Summary cards	Auto-generated from the summary dict keys
Bar chart	For Burn Rate and Daily reports
Preview table	Styled rows with status-color cells

8.2.5 ? Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
? Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> ? Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st 06:00
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

- Report icon + label

- Freque

8.4 Reports -> AI Insights

If AI_ENABLED and Ollama is running with llama3.1:8b:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 fixed SQL probes + LLM commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	LLM-reported confidence
Right-side metric callout	Headline number with sub-label
Body paragraph	LLM-generated explanation grounded in the SQL probe result
? Recommendations list	Up to 3 numbered actions
? Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 fixed probes:

1. Abnormal consumption spike (vs trailing average)

2. Items a

If Ollama is not running: a graceful fallback explains setup steps.

8.5 Reports -> ? Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download

- ? Re-e

9. Automated Notifications - WhatsApp & Email

The system automatically queues messages on key events. Background worker (`whatsapp_worker.py`) drains the queue.

9.1 WhatsApp triggers (auto-fire)

Event	Recipient	Trigger location	Notes
Store Keeper submits issue batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits receipt batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits stock count	Site HOD	Entry Log Stock Count Submit	Includes variance + reason
Store Keeper performs FEFO override	Site HOD	Entry Log Add to Grid	Includes chosen vs FEFO-suggested lot +
HOD commits EOD	Site HOD (self)	post-commit	Only if low-stock items result
HOD cross-site request created	All Admins	Submit cart	Includes target site + count
HOD cross-site request, > 5 items	Target Site HOD	Submit cart	Bulk escalation
Admin approves cross-site requests (batch)	Requesting HOD	Approve Selected	Includes approved items + admin notes
Admin approves cross-site requests (batch)	Target Site HOD	Approve Selected	"TRANSFER ORDER - pack and ship"
Admin rejects cross-site requests (batch)	Requesting HOD	Reject Selected	Includes rejection reason
Returnable item overdue	Borrower + Store Keeper	Background scheduler	Time-driven, not in current scope
New self-registration request	All Admins	Registration form	Includes username + role + site
Self-registration approved	Requesting user	Admin User Mgmt	"ACCESS GRANTED - Welcome"
Manual send	Free	HOD Notifications / Admin WhatsApp Console	Audit MANUAL_WHATSAPP

9.2 Email triggers

Event	Trigger location	Mechanism
Logistics PR follow-up	HOD PR tab -> ? Draft Outlook Email	Mac: AppleScript-driven Mail.app with HTML table. Windows: COM Outlook with HTMLBody. Fallback: mailto: with
EOD report email	mailer.py SMTP	Configure plain-text table (SMTP_SERVER, SMTP_USER,
Report delivery (scheduled)	Reports -> Scheduled	Same SMTP pipeline

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (PK), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override,	? subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current_Stock} = \text{Total_Received} - \text{Total_Consumed} + \text{Total_Returned}$ - computed live in v_site_stock view.

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	API tokens for offline PWA
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	? Damaged / unusable	Physical damage
expired_disposal	? Expired - disposed	Expired and physically discarded
miscount_in	? Miscount - found extra	Found more than system shows
miscount_out	? Miscount - short	Found less than system shows
lost	? Lost / unaccounted	Genuinely missing
theft	? Suspected theft	Deliberate removal
return_to_supplier	<-? Returned to supplier	Quality issue, sent back
other	? Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent

- submit

11.4 Lot statuses

- open - has remaining quantity, available for FEFO

- exhaust

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	Write-Ahead Logging (SQLite concurrency mode)
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> ? Access Control -> ? Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

13. 2026-06 Feature Update - What Changed

This section documents the upgrades shipped in the 2026-06 release. Everything above remains accurate except where this section overrides it.

13.1 Field-level changes

Change	Where	Behaviour
All form fields mandatory	Entry Log (Consumption / Receipt Staging), HOD Receive Material, Admin Add Entry forms	Every text/number/select input is required. Validation lists missing fields on submit.
Expiry Date is optional	SK Receipt Staging, HOD Receive Material	Marked (Optional). Leave blank for non-perishable items.
Remarks, Tank No., Serial No., PR Number	Same forms	No longer optional.

13.2 Live Dashboard column order

The Live Dashboard table now renders in this order (when columns exist):

SAP_Code -> Material_Code -> Equipment_Description -> UOM -> Opening_Stock -> Receipt -> Consumption -> Return -> Closing_Stock -> Minimum_Qty -> Unit_Cost -> Stock_Value -> Category -> Status

- Opening_Stock is now a configurable column on inventory. Default 0; admin can edit in DB Editor.

- Identity
Total_I

13.3 Entry Log access - Store Keeper only

The Entry Log page is now visible **only to the `store\_keeper` role**. HODs review submissions in HOD Portal; Admins use Admin Portal. The page is hidden in the sidebar for other roles.

13.4 EOD Commit - checkbox confirmation

The "Confirm EOD Commit" dialog no longer requires typing **COMMIT**. Tick the confirmation checkbox and click **Confirm Commit**. Cancel still drops all pending state.

13.5 Material Category

Every inventory item now carries a **Category**. Categories: **Consumable**, **Equipments**, **Utilities**, **Maint**

enance, Others (default), Rubber materials, Tools, QC items.

- Admin Portal Add New Entry -> renders Category as a selectbox.

- Report Category

13.6 Rubber MTC workflow

When a Store Keeper stages a receipt and the selected material's category is **Rubber materials**, the system shows:

- MTC Number text field (e.g. MTC-2026-001234)

- MTC D

Either field can be blank - the receipt still goes through. What changes is HOD-side visibility:

- HOD Portal -> Pending Receipts: a red banner lists rubber items received without an MTC.

- Click D qty.

The logistics email recipient defaults to `LOGISTICS_EMAIL` env var (`logistics@generalindustries.net` if unset). On Windows it opens Outlook; on macOS it opens Mail.app; on Linux it opens the default mailto handler.

13.7 Document attachments - Entry Log + HOD DOC tab

Store Keepers can attach reference documents (PDF / JPEG / JPG / XLSX) on:

SK form	Doc number used	Notes
Consumption Log	Auto = DDMMYY of the date	Pick scope: "Whole entry (batch)" or "Specific date".
Receipt Staging	DN No. of the row (or manual override)	Falls back to DN-DDMMYY if no DN_No found.

Files are stored **as BLOBs inside the database** (authoritative copy) and **mirrored to** `uploads/<Site_ID>/<doc_type>/<doc_number>/` for local browsing. The disk mirror is gitignored; only the DB BLOB is portable.

HOD Portal -> DOC is a new tab with three sub-tabs: **Consumption** / **Receipt** / **<? Return**. Each shows period (From/To dates) and Doc Number text filters, with a per-file ?? download button.

The **<? Return** sub-tab pulls from the new Return Items workflow (see §13.10), not from Returnable Items.

13.8 QR Label approval flow

The Admin DB Editor's QR generator (single-user, single-item) is unchanged. The new flow is two-step:

1. Store Keeper -> Entry Log -> QR Label Request (new tab)

- Multi-s

1. HOD Portal -> QR Approval (new tab)

- ? Penco

13.9 Returnable Items - clarification

The **Returnable Items** tab is for **temporary tool loans only** (e.g. issuing a torque wrench to a worker who'll return it before EOD). It is *not* a way to return stock to the warehouse. No DN No., no document attachment.

For real returns (defective material going back to logistics), use the new Return Items tab - see §13.10.

13.10 Return Items workflow (NEW)

The new **<-? Return Items** tab (between Receipt Staging and Returnable Items in the Entry Log) handles real returns to the warehouse / logistics.

Store Keeper flow

1. The material picker is restricted to materials received in the last 30 days at the user's site.

2. If multi
(Date /

HOD flow

1. HOD Portal -> <-? Returns lists every pending return with a card per row.

2. Rows t

Dashboard / report impact

- The Return column on the Live Dashboard reflects approved returns (since returns is the source of truth).

- All existing returns are the same as the source of truth.

13.11 Per-site Work Type and Tank No.

HOD Portal -> **Site Config** lets the site HOD add or delete Work Types and Tank Numbers scoped to their site. Empty per-site lists fall back to the global defaults.

13.12 WhatsApp worker - startup fix

The worker module no longer imports `pywhatkit` at module load (that pulled in heavy GUI deps and stalled local launch by tens of seconds). It now lazily imports `pywhatkit` only when an outbound message has no Twilio fallback. On Streamlit Cloud, Twilio handles delivery; on local desktop, `pywhatkit` is loaded the first time a message is queued.

If you see the spinner sitting on `_start_whatsapp_worker()` for more than ~3 seconds locally, check that you ran `pip install -r requirements.txt` after the update.

Document end

This manual covers every page, tab, button, table, and field built into the General Industries Hub v2.0 as of the latest commit. For technical reference (function signatures, table schemas, full SQL), see `database.py`, `auth.py`, and `pages_internal/*` source files.

For PDF export: use any markdown-to-PDF converter (Typora, pandoc, marp, or VSCode's "Markdown PDF" extension). Suggested pandoc command:

```
pandoc USER_MANUAL.md -o USER_MANUAL.pdf --pdf-engine=xelatex \
  --toc --number-sections \
  -V geometry:margin=2cm -V mainfont="Helvetica" -V monofont="Menlo"
```